

CHE 201 Spring 2020 HW 4

Anna Christensen

TOTAL POINTS

84 / 100

QUESTION 1

1 Question 1 15 / 20

- 0 pts Correct
- ✓ - 5 pts "...in terms of their individual thermal efficiencies"

- 2.5 pts Insufficient component about the phase-out schedule
- 5 pts Insufficient component about the replacer
- 2.5 pts Insufficient component about the replacer
- 1 pts No enough references

QUESTION 2

2 Question 2 15 / 20

- 0 pts Correct
- 5 pts Wrong V2
- 5 pts Wrong work
- ✓ - 5 pts Wrong internal energy difference
- 5 pts Wrong final temperature
- 20 pts Did not answer 2.53

QUESTION 3

3 Question 3 20 / 20

- ✓ - 0 pts Correct

QUESTION 4

4 Question 4 14 / 20

- 0 pts Correct
- 20 pts Did not answer problem 3.7
- ✓ - 3 pts (a) Wrong sketch of T-v diagram
- 3 pts (a) Wrong specific volume
- 3 pts (b) Wrong sketch of T-v diagram
- ✓ - 3 pts (b) Wrong specific volume
- 3 pts (c) Wrong sketch of T-v diagram
- 3 pts (c) Wrong specific volume

QUESTION 5

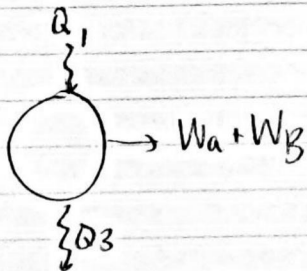
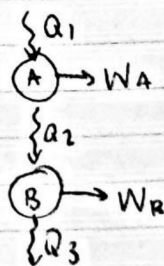
5 Question 5 20 / 20

- ✓ - 0 pts Correct
- 20 pts Did not answer 3.4D
- 5 pts Insufficient component about the phase-out schedule

HW 4

2/16/2020

1 ①



$$M = \frac{W_{\text{cycle}}}{Q_{\text{in}}}$$

$$M_A = \frac{W_A}{Q_1}$$

$$M_B = \frac{W_B}{Q_2}$$

$$M_{\text{total}} = \frac{W_A + W_B}{Q_1}$$

②

$$P_1 = 100 \frac{\text{lbF}}{\text{in}^2}$$

$$V_1 = 3.704 \text{ ft}^3/\text{lb}$$

$$T_1 = 1000^\circ\text{R}$$

$$P_2 = 50 \frac{\text{lbF}}{\text{in}^2} \quad \gamma = 1.4$$

$$\Delta u = (0.171)(T_2 - T_1) \frac{\text{Btu}}{\text{lb}}$$

* find T_2

$$P V^{1.4} = \text{constant}$$

$$P_1 V_1^{1.4} = P_2 V_2^{1.4}$$

$$W = \int P dV$$

$$W = \Delta u$$

$$100 \frac{\text{lbF}}{\text{in}^2} (3.704)^{1.4} = (50 \frac{\text{lbF}}{\text{in}^2}) V_2^{1.4}$$

$$V_2 = 6.077 \text{ ft}^3/\text{lb}$$

$$\int P dV = (0.171)(T_2 - T_1)$$

$$P V^{1.4} = \text{constant}$$

$$P = \frac{\text{constant}}{V^{1.4}}$$

$$P_1 V_1^{1.4} = 100 (3.704)^{1.4}$$

$$\text{constant} = 625.4$$

$$625.4 \cdot \int_{V_1}^{V_2} \frac{1}{V^{1.4}} dV = (0.171)(T_2 - 1000^\circ\text{R})$$

$$625.4 \left(-\frac{5}{2} V^{-2/5} \right) \Big|_{3.704}^{6.077} = (0.171)(T_2 - 1000^\circ\text{R})$$

$$625.4 (0.246) = (0.171)(T_2 - 1000^\circ\text{R})$$

$$T_2 = 1972.8^\circ\text{R}$$

1 Question 1 15 / 20

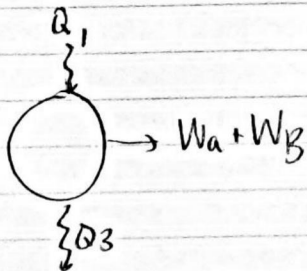
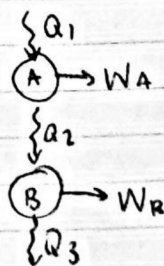
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HW 4

2/16/2020

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2 Question 2 15 / 20

- 0 pts Correct
- 5 pts Wrong V2
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- ✓ - 5 pts Wrong internal energy difference
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67 (3) power input: 0.15 kW
 output: 0.6 kW
 Q_{out}

Determine at rate in which
 energy is removed by heat
 transfer
 + coefficient of performance

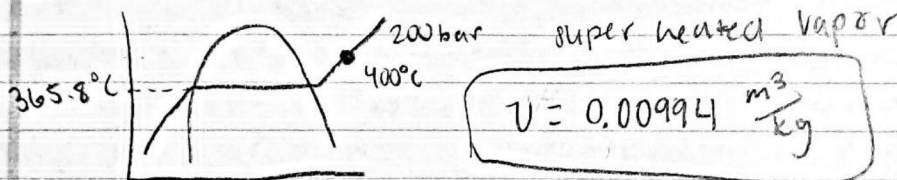
$$W = Q_{out} - Q_{in}$$

$$Q_{in} = Q_{out} - W$$

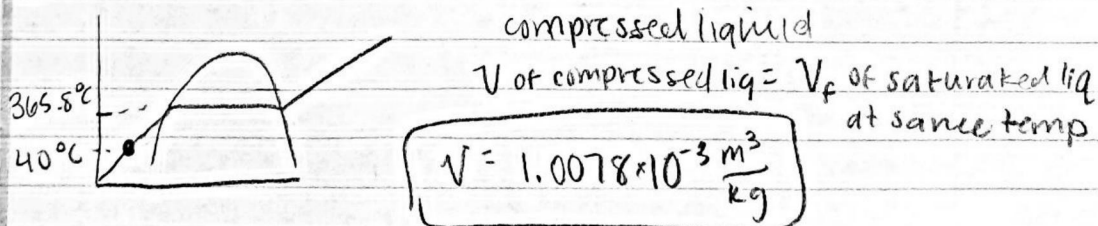
$$Q_{in} = 0.6 - 0.15 = 0.45 \text{ kW}$$

$$\text{coefficient of performance} = \beta = \frac{Q_{in}}{W_{cycle}} = \frac{0.45}{0.15} = 3$$

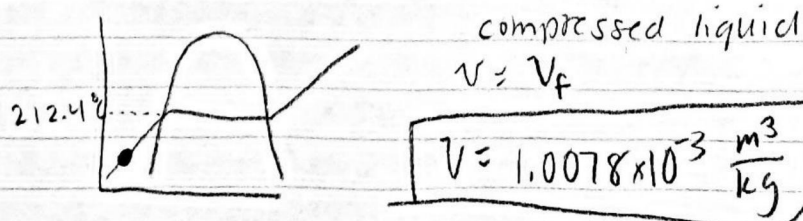
7 (4) Determine specific volume in $\frac{\text{m}^3}{\text{kg}}$ locate states on T-V diagrams
 (a) $T = 400^\circ\text{C}$ $p = 20 \text{ MPa} = 200 \text{ bar}$



(b) $T = 40^\circ\text{C}$ $p = 20 \text{ MPa} = 200 \text{ bar}$



(c) $T = 40^\circ\text{C}$ $p = 2 \text{ MPa} = 20 \text{ bar}$



3 Question 3 20 / 20

✓ - 0 pts Correct

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 output: 0.6 kW
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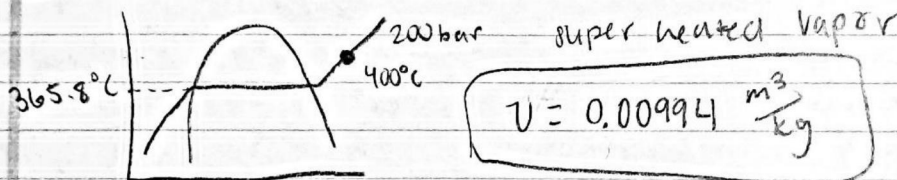
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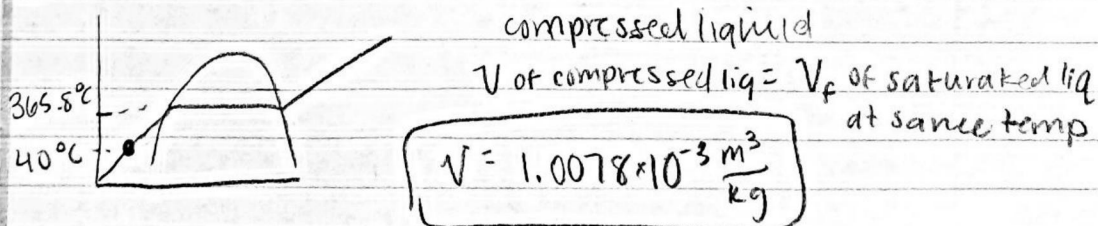
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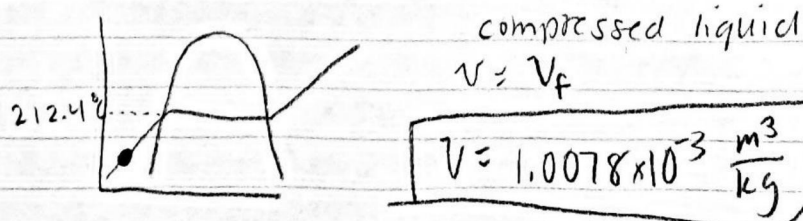
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The U.S. Environmental Protection Agency (EPA) has set January 1, 2022 as the date when R-22 is banned from production and importation in the United States. R-22 that is still in use will not be affected. HVAC systems that use R-22 will be obsolete. If a system that uses R-22 needs a repair that requires adding more R-22, then the only R-22 that can be added is the previously produced R-22. This means that the cost of refrigerant will rise dramatically.

(<https://www.air2o.com/why-should-you-care-about-the-r22-phase-out/>)

For any appliance, the EPA maintains a list of refrigerants that can be used as alternatives for R-22. However, any alternative will not work on all appliances without making modifications to the systems. When switching over the refrigerants, people should recover the old refrigerant in order to protect the ozone. It is illegal to release refrigerant.

Alternatives to R-22 that do not cause depletion to the ozone layer are R-134a, R-507, R-404A, R-407c, and R-410A. For air condition appliances the best choice is R-410A and for refrigeration the best choice is R-404A and R-507. The most common alternative is R-410A which is a non-ozone-depleting HFC refrigerant blend.

(<https://www.brighthubengineering.com/hvac/62099-alternative-refrigerants-for-r22/>)

The U.S. phaseout had reduced the amount of ozone-depleting substance that was legally produced or imported by doing it in stages. The Parties to the Montreal Protocol have changed the phaseout schedule over time. The EPA has accelerated the phaseout in the U.S. under its Clean Air Act authority. The schedule started January 1, 2010, when there was a ban on production, import and use of HCFC-22 and HCFC-142b, except for continuing servicing needs of existing equipment. The next stage was January 1, 2015, where there was a ban on all HCFCs, except for continuing servicing of existing equipment. Then finally, on January 1, 2020, they ended the continuing services of existing equipment. All existing equipment after this date will rely on recycled or stockpiled quantities of R-22. The final stage will take place January 1, 2030, where the ban will include all remaining production and import of all HCFCs. The sectioned stages of the phase out allowed people to make updates in their equipment or to purchase new equipment before they completely ban the production of R-22 which would not allow new service to be done on the equipment. (https://www.epa.gov/sites/production/files/2015-07/documents/phasing_out_hcfc_refrigerants_to_protect_the_ozone_layer.pdf)

5 Question 5 20 / 20

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- 20 pts Did not answer 3.4D
- 5 pts Insufficient component about the phase-out schedule
- 2.5 pts Insufficient component about the phase-out schedule
- 5 pts Insufficient component about the replacer
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